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Further palaeomagnetic data from Chitral (Eastern Hindukush): evidence for an early India–Asia contact

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Abstract

The Eastern Hindukush forms part of an elongate belt ("Central Domain", collage of Cimmerian microcontinents) that encircles the northern part of the Indian subcontinent. A Gondwanan origin is commonly assumed for this belt, but a "Laurasian" origin for the Chitral region has been argued on palaeontological (Talent and Mawson, 1979) and palaeomagnetic (Klootwijk and Conaghan, 1979) grounds. The "Laurasian" view was based on a pilot study we undertook of Upper Devonian pisolitic ironstones from a thrust sheet at Kurāgh Spur in Chitral. Preliminary results showed a characteristic magnetization component [$D = 318°$, $I = -6.5°$, $N = 7$ (block samples), $k = 14$, $\alpha_{95} = 16.5°$] indicating an equatorial palaeoposition. This component was thought to be of primary origin and was interpreted in terms of a Late Devonian "Laurasian" affinity of the Kurāgh Spur rocks. This controversial conclusion has been tested in the present more comprehensive study of the thrust pile of sedimentary rocks in the Reshūn–Kurāgh–Būni region of Chitral and the primary origin of the characteristic magnetization component refuted.

Thermal demagnetization of 333 block samples from Middle to Upper Devonian variegated sediments, Permian quartz flysch, Permo-Triassic carbonates, and mid-Cretaceous redbeds showed two interpretable components. A softer component of recent origin (A); and a harder characteristic component (B) of both normal and reverse polarity whose mean direction [$D = 314.1°$, $I = 6.0°$, $N = 4$ (thrust sheets), $k = 198.2$, $\alpha_{95} = 6.5°$] is comparable to the characteristic component observed in our preliminary study. However, the universal presence of this component throughout the thrust pile proves its overprint origin, which we attribute to initial India–Asia contact. Palaeomagnetic information pertinent to the controversy of a "Laurasian" versus a Gondwanan origin of the Chitral region has not been obtained in this further study because primary magnetizations could not be identified beyond doubt. Hence, we retract herewith our original conclusion of a Late Devonian "Laurasian" affinity of the Chitral region on the basis of the palaeomagnetic evidence.

The secondary component (B) comprises a suite of secondary magnetizations, acquired at equatorial-to-low-northern palaeolatitudes, and is attributed to initial contact between Greater India and southern Asia. Component B has been observed previously in the Himalayan–Tibetan region, both north and south of the Indus–Tsangpo

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Suture zone. Identification is herein extended to the Hindukush region north of the Northern Kohistan (or Shyok) Suture zone, which is a western continuation of the Indus–Tsangpo Suture. Comparison of this suite of collision-attributed equatorial palaeolatitude data from the India–Asia convergence zone with new palaeolatitude constraints from the Ninetyeast Ridge on the northward movement of the Indian plate, constrained additionally by a recent minimal estimate of the palaeogeographic northern extent of Greater India, indicates that initial contact between northwestern Greater India and southern Asia was established at, or before, the Cretaceous–Tertiary boundary. The overprint origin of component B at about this time is further supported by observations by Zeitler (1985) on rocks from the sampled area in Chitral of partially reset zircon fission-track ages around 68–55 Ma.

The NW–SE declination axis of component B indicates a 60–70° counterclockwise rotation of the sampled thrust pile with respect to Eurasia and a counterclockwise rotation between 10 and 30° with respect to India. Some of the recent field components (A) show a comparable rotation and indicate that the tectonic activity that led to the formation of the Hindukush–Pamir–Karakorum syntaxial zone has continued into recent times.

1. Introduction

This study is a follow-up to a palaeomagnetic reconnaissance (Klootwijk and Conaghan, 1979) of Upper Devonian sedimentary ironstones from Chitral (Pakistan) in the Eastern Hindukush. The results of the earlier study of a limited collection of sedimentary ironstones showed a characteristic magnetization component that we interpreted as having been primary and indicative of an equatorial palaeolatitude. Comparison with the then available palaeomagnetic data from "Laurasia" and Gondwana together with palaeontological arguments (Talent and Mawson, 1979) led us to conclude that the Chitral region is an original part of Asia and does not have Gondwanan affinity, but also that the Eastern Hindukush underwent large-scale counterclockwise rotation and northward displacement relative to the Siberian platform as a result of India's penetration into southern Asia during the Cainozoic.

The present study of an extensive collection of Middle Devonian–Cretaceous sedimentary rocks from Chitral, including additional samples of the previously studied ironstones from Kurāgh, confirms the earlier observed characteristic magnetization. However, the persistent presence of this characteristic component throughout different sampled sedimentary successions in several thrust sheets proves its secondary origin. Consequently this component has no bearing on the original supercontinent affinity of the Chitral region.

Secondary magnetization components with equatorial palaeolatitudes comparable to this characteristic component have been observed in other studies throughout the wider Himalayan–Tibetan region, and have been interpreted as having been acquired upon initial India–Asia contact. Their comparison with new palaeolatitude control on India's northward motion during the past 80 Myr, obtained from ODP Leg 121 studies on Ninetyeast Ridge (Klootwijk et al., 1991, 1992), constrains the timing and the location of initial India–Asia contact. The present study focuses more on this comparison than on analysis of the "Laurasian"-versus-Gondwanan affinity of the Chitral region as originally intended.

2. Regional geological framework

Studies of south-central Asia's Phanerozoic development (e.g., Stöcklin, 1977, 1989; Bouyx, 1981; Boulin, 1981, 1988, 1991; Helmcke, 1985; Şengör, 1987; Chang Chen-fa et al., 1986, 1989; Smith, 1988; Girardeau et al., 1989; Zonenshain et al., 1990; Searle, 1991 and references therein; Le Pichon et al., 1992; Metcalfe, 1993) emphasize its episodic growth through accretion of various microcontinents and microcontinental belts prior to terminal collision of Gondwanan India with Eurasia (Fig. 1). South of the older of these belts and sutures peripheral to the Siberian block, three successively younger sutures have been identified peripheral to the Indian subcontinent. These can be followed with varying degrees of continuity eastwards into Southeast Asia and westwards through Iran into Turkey (Fig. 1).

These younger peri-Indian sutures are be-

lieved to relate to the closure of Tethys, but the manner in which this occurred is much disputed. Some (e.g., Stöcklin, 1989) argue against the pre-Triassic existence of the large eastward-opening Tethyan gap commonly shown on reconstructions of Pangea and see this gap as an artefact of earth expansion. Most relate the two northernmost peri-Indian sutures to closure of the Palaeotethys, but differ on the number of accreted belts and the timing of the process. Extremes range from

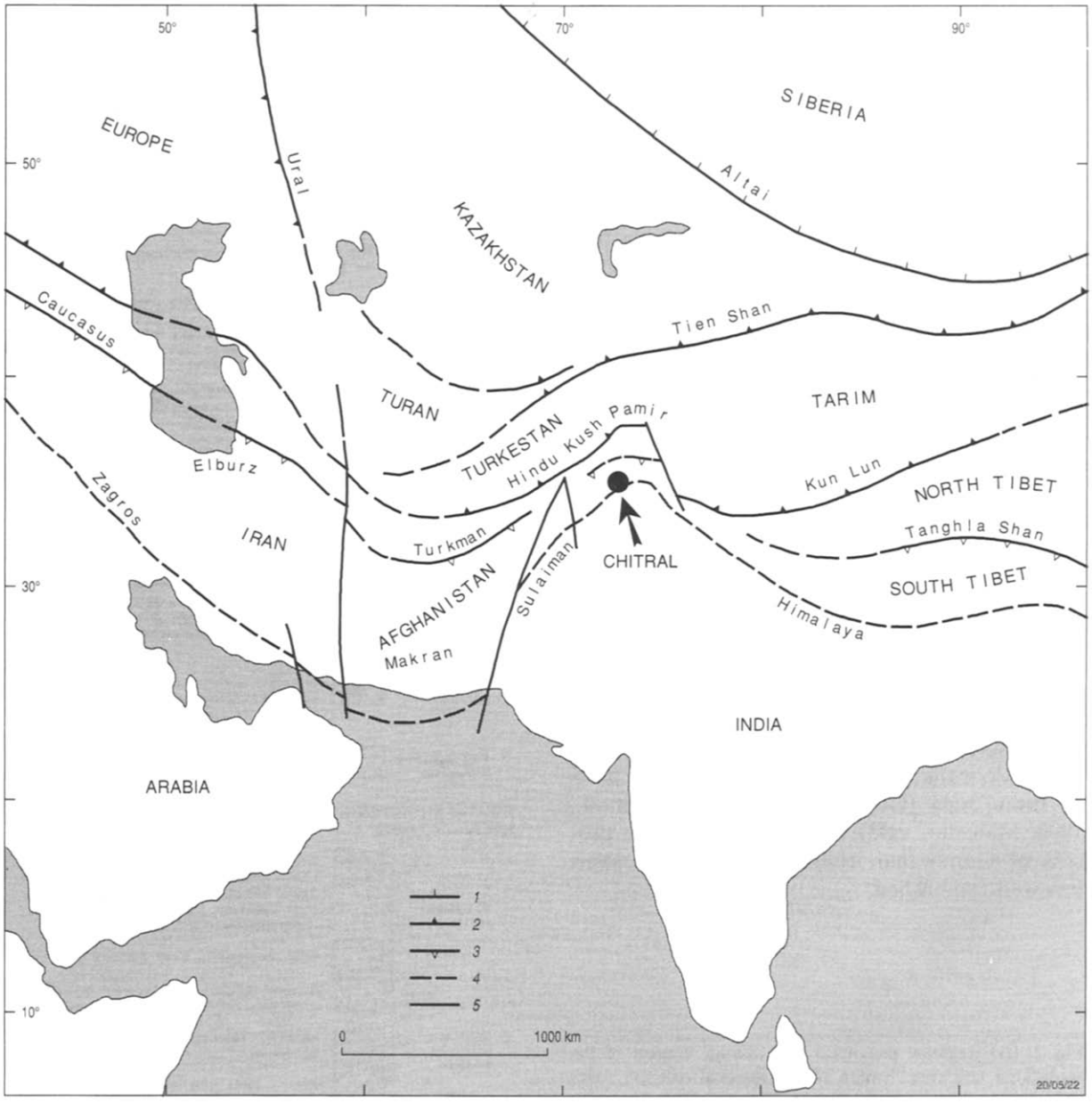


Fig. 1. Main ophiolitic suture zones of central and southern Asia after Boulin (1981, fig. 3). *1* = Caledonian? suture; *2* = Hercynian suture; *3* = Cimmerian suture; *4* = Alpine sutures; *5* = regional faults.

numerous terranes (Chang Chen-fa et al., 1986) to a single Cimmerian continent (Şengör, 1987; Şengör et al., 1988), with the timing of closure ranging between the Late Devonian–Carboniferous (Boulin and Bouyx, 1978; Boulin, 1981; Helmcke, 1985) and the Late Triassic–Early Cretaceous (Şengör, 1987; Şengör et al., 1988).

Most relate the third, southernmost, peri-Indian suture to closure of the Neotethys with opinions on the timing of initial contact between Greater India and Eurasia varying from the Cretaceous–Tertiary boundary (KTB) (Jaeger et al., 1989; Klootwijk et al., 1991) to various times in the Tertiary [Dewey et al., 1989 (45 Ma); Patriat and Achache, 1984 (55 Ma)]. Closure of this ocean and subsequent indentation of India into southern Asia (Tapponnier et al., 1982) and intracontinental convergence of India across the Himalaya (Powell and Conaghan, 1973; Klootwijk et al., 1985) have led to major reorientation of the inferred microcontinental accretion zones around India's periphery, reflected in the Hindukush–Pamir–Karakorum curvature (or "Pamir knot", Figs. 1, 2A, 11).

The fusion of Gondwanan continental blocks to southern Eurasia is now reliably constrained for: (1) regions in Turkey, Iran and Afghanistan west of the Chaman Fault–Sulaiman mobile zone (Figs. 1, 2A) that defines the western limit of the Indian continental block (Dercourt et al., 1986; Şengör, 1987; Şengör et al., 1988; Searle, 1991); (2) the Indian continental block north to its exposed limit along the southernmost of the peri-Indian sutures (Figs. 1, 2A; Patriat and Achache, 1984; Klootwijk et al., 1985, 1991); and (3) parts of Indo-China (Helmcke, 1985; Burrett et al., 1990; Metcalfe, 1993). Elsewhere the fusion process of Gondwanan continental blocks remains less well established.

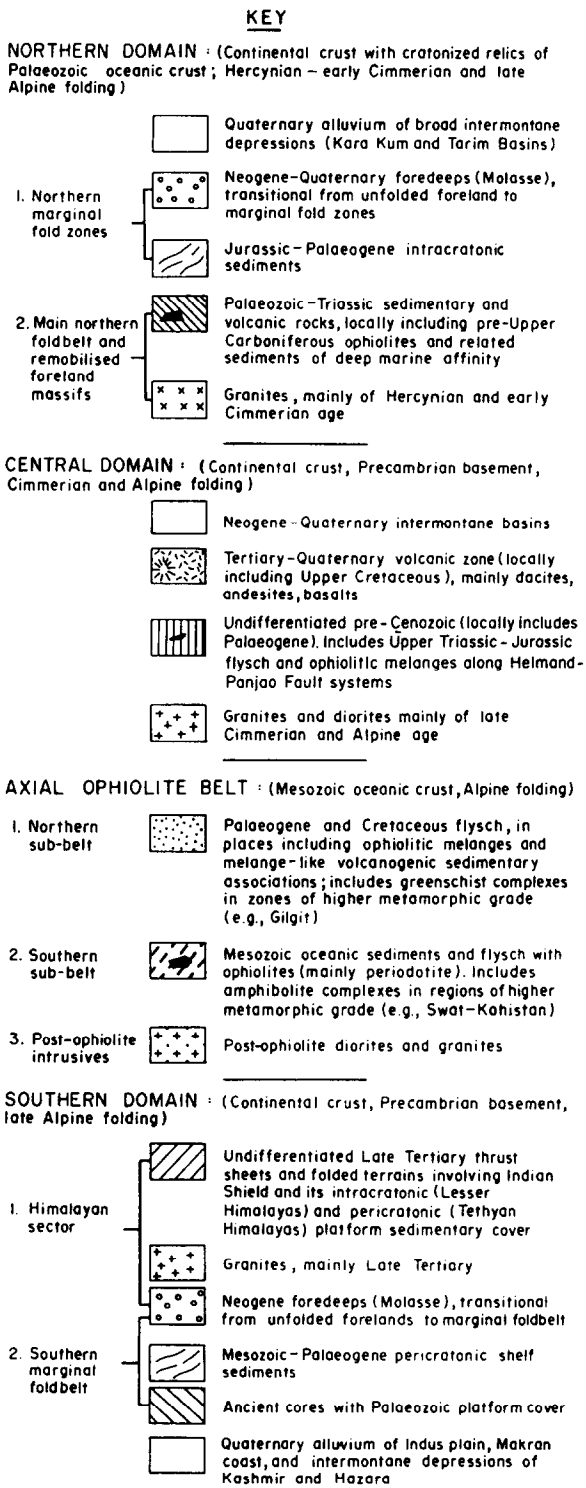
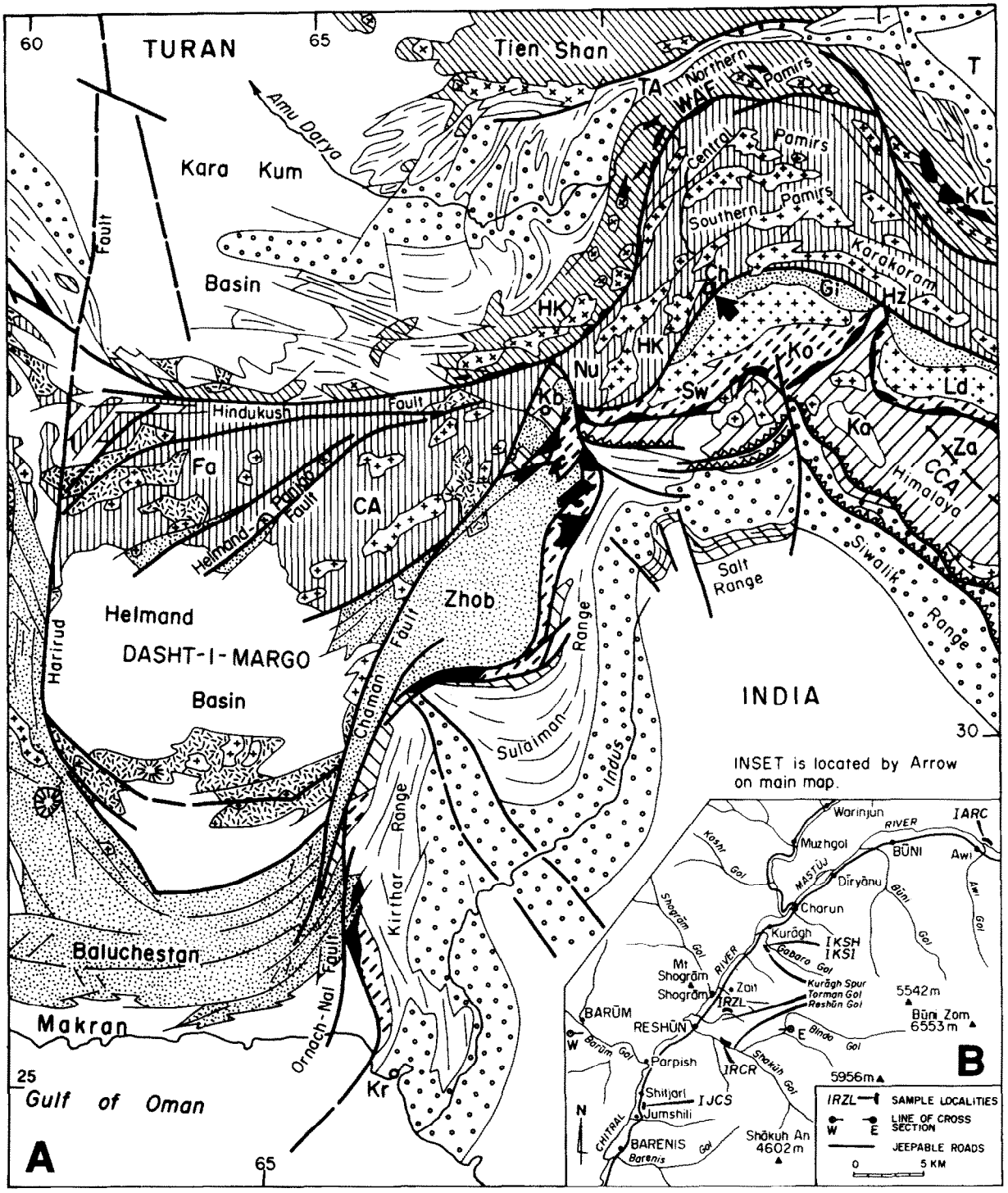


Fig. 2. (A) Regional geological and tectonic context of the study area and classification of the regional tectonic zones according to Stöcklin (1977, fig. 1). The study area (arrow) is located towards the top-right corner of the map. (B) Location map of sample localities and the cross section shown in Fig. 3.



Ca Central Afghanistan
CCA Central Crystalline Axis of Himalaya
Ch Chitral
Fa Farah zone
Gi Gilgit
HK Hindukush
Hz Hunza
Ka Kashmir
Kb Kabul
Ko Kohistan
KL Kun - Lun
Kr Karachi
Ld Ladakh
Nu Nuristan
Sw Swat
T Tarim Basin
Ta Transalai
WAF Wanch - Akbaytal Fault
Za Zanskar

Fig. 2 (continued).

3. First-order tectonic elements: the "Central Domain"

The geological integrity and Gondwanan affinity of the mosaic of microcontinents between the northern and southern peri-Indian sutures in the region west of the Chaman Fault were first emphasized by Stöcklin (1977) and Şengör (1987) as part of a Tethyan-wide Cimmerian continent or collage of microcontinents. This recognition is a key element in the tectonic classification of the wider region (Fig. 2A), wherein the zone of microcontinents comprises a "Central Domain" separating Laurasia/palaeo-Asia ("Northern Domain") from Gondwanan India–Afro-Arabia ("Southern Domain") and its flanking Zagros–Baluchestan–Kohistan–Indus–Tsangpo Suture ("Axial Ophiolite Belt"). These first-order tectonic subdivisions were distinguished by Stöcklin and Şengör, primarily on the basis of fundamental differences in crustal character and chronology of basement consolidation (Fig. 2A).

At Kabul the "Central Domain" is partitioned structurally into western and eastern sectors (Fig. 2A) which Stöcklin (1977) argues reveal a common geological integrity. Whilst available geological, faunal and palaeomagnetic evidence from the western sector is consistent with a Gondwanan origin, this origin is less well established for the eastern sector. Various palaeobiogeographic commentaries (Bassoullet et al., 1980; Helmcke, 1985; Chang Chen-fa et al., 1986; Metcalfe, 1993), stressing correlations with Southeast Asia, favour a Gondwanan origin; but Talent and Mawson (1979) and Talent et al. (1982) have argued on the basis of palaeobiogeographic evidence that the Chitral–Gilgit–Hunza region of the Eastern Hindukush is not a former part of Gondwana. Talent et al. (1987), while accepting that most of the microcontinents comprising Stöcklin's "Central Domain" are of Gondwanan origin, still have reservations regarding the timing of accretion to Laurussia/Laurasia of their "Pamir–Karakorum block". This latter, possibly composite, block (J.A. Talent, pers. commun., 1993), includes the Chitral–Hunza area. There are presently no decisive palaeomagnetic data available from the eastern sector to substantiate its inferred Gondwanan affinity.

4. Palaeomagnetic evidence: the eastern sector of the "Central Domain"

Palaeomagnetic data from regions of the postulated microcontinental accretion zone north of the southernmost peri-Indian suture do not shed a decisive light on the "Laurasian"-versus-Gondwanan original affinity of this zone. The data are either from strata that postdate the postulated episodes of fragmentation and accretion or are of a suspect nature in view of widespread collision-related magnetic overprinting (Zhu Zhi Wen et al., 1981; Pozzi et al., 1982; Westphal et al., 1983; Achache et al., 1984; Lin and Watts, 1988; Otofuji et al., 1989).

Our earlier study (Klootwijk and Conaghan, 1979) of Upper Devonian sedimentary ironstones from Kurāgh, Chitral was based on a small number of samples collected by P.J.C. in the course of stratigraphic work in Chitral during 1975. This material yielded a characteristic palaeomagnetic direction [$D = 318°$, $I = -6.5°$, $N = 7$ (block samples), $k = 14$, $\alpha_{95} = 16.5°$] that we interpreted to be of primary origin, notwithstanding the tectonically disturbed nature of the Chitral region. These results indicated a palaeoposition for the Chitral Devonian rocks immediately south of the equator (3°S, $dp = 8.5°$, $dm = 16.5°$). Whilst this position was not in agreement with the then expected Late Devonian palaeoposition of Chitral under the assumption of either a Gondwanan origin ($\sim 45°$S) or a "Laurasian" origin (20–30°N), we opted for a pre-Late Devonian "Laurasian" affinity in keeping with palaeobiogeographic arguments (Talent and Mawson, 1979; Talent et al., 1982). The difference between the observed and expected palaeomagnetic results we attributed to large-scale crustal shortening and counterclockwise rotation of the area (as part of the western limb of the Pamir knot) in response to India's Cainozoic indentation of southern Asia. The palaeomagnetic data presented in the present paper were sought to confirm and extend these

results and stem from a more comprehensive sampling program in Chitral by C.T.K and R.N. during 1977, guided by P.J.C.'s stratigraphic mapping in the area in 1973 and 1975.

At the time of our earlier study, interpretation of the characteristic magnetization component of the Kurāgh rocks as being of primary origin seemed justified because of: (1) the general absence of penetrative deformation throughout most of the Upper Devonian succession, and absence of evidence of metamorphism as witnessed additionally by an abundance of undeformed, not-recrystallized shelly fossils; (2) the primary sedimentary origin that can be concluded for the chamositic ironstones; (3) the antipodal normal and reverse polarity directions that characterize the palaeomagnetic pattern; and (4) the magnetic hardness of the observed components with blocking temperatures ranging from 300 to 600°C. Evidence of organic metamorphism (destruction of plant microfossils) in nearby tectonically juxtaposed Cretaceous sediments (W.K. Harris, written commun. to P.J.C., 1974) indicated the probable heating of the region to temperatures above 250°C, but we did not consider this sufficient reason to negate interpretation of the magnetic components as primary. However, subsequent conodont colour-alteration studies (Molloy, 1979; written commun. to P.J.C., 1981) of material from the Devonian succession at Kurāgh and adjacent localities, and of conodont material from Triassic rocks nearby, indicated temperatures up to and in excess of 300°C [colour-alteration index (CAI) ranges from 3.5 to 6], subsequently confirmed by Zeitler (1985) on the basis of fission-track cooling-age studies of rocks of the Chitral area. Moreover, with the progress of additional palaeomagnetic studies in other regions of the peri-Indian mountain belts, it became increasingly evident that the equatorial palaeolatitude indicated by our earlier Chitral palaeomagnetic study is very similar to that of secondary magnetization components observed in rocks of various ages in Ladakh (Klootwijk et al., 1979; Klootwijk, 1984), Nepal (Klootwijk and Bingham, 1980; Gautam and Yoshida, 1988), southern Tibet (Zhu Zhi Wen et al., 1981), and northwestern Pakistan (Klootwijk et al., 1981, 1991; Haag and Heller, 1991). Acquisition of these secondary components was attributed to the India–Asia collision process (Klootwijk, 1984; Klootwijk et al., 1985), whose timing and evolution have been a matter of extensive debate. Comparison of these secondary magnetizations with the northward movement of Greater India, as detailed recently from ODP Leg 121 studies on the Ninetyeast Ridge (Klootwijk et al., 1991, 1992), provides crucial control on the timing and the location of initial India–Asia contact.

We document here the new palaeomagnetic data from Chitral, show their overprint nature, revise our earlier interpretation of the preliminary dataset, and interpret the new data in terms of timing of initial India–Asia contact.

5. Geological setting of the study area

The area studied occupies a 40-km-long section of the valley of the Chitral–Mastūj River in the Chitral District of the North West Frontier Province and, orographically, lies towards the eastern extremity of the Eastern Hindukush (Fig. 2). In terms of Stöcklin's tectonic subdivision of the wider region, it lies within the eastern sector of the "Central Domain", but adjacent to the Axial Ophiolite Belt that delineates its southern boundary (Fig. 2A). This places the area in the southern of the two speculative microcontinental accretion belts that subsequent workers recognize on the basis of evidence in Afghanistan, Tibet and Southeast Asia (viz., *northern*: Turkman–Farah Rud–Chang Tang–Indochina belt; *southern*: central Afghanistan–Nuristan–southern Tibet–Sibumasu belt; Fig. 1; e.g., Bassoullet et al., 1980; Bouyx, 1981; Boulin, 1981, 1988; Helmcke 1985; Şengör, 1987; Smith, 1988; Girardeau et al., 1989; Metcalfe, 1993), but which cannot be differentiated in the intervening region of the Pamir knot because of impersistence or apparent loss of the defining Cimmerian suture across this syntaxis (Figs. 1, 2A). In this scenario, amalgamation of the southern microcontinental belt with an already coherent Eurasian landmass to its north is believed to have occurred during the interval from the Late Jurassic to the Early Cretaceous

(Bassoullet et al., 1980; Boulin, 1981, 1988; Chang Chen-fa et al., 1989; Girardeau et al., 1989), as witnessed by widespread contemporary (Late Cimmerian) deformation and plutonism throughout the "Central Domain". These events were then overprinted by the intense Alpine tectonism and associated plutonism that accompanied Greater India's collision with and indentation of southern Asia, resulting in formation of the northward-convex Pamir knot, of whose western limb the study area is a part.

Regional structural trends in the study area are aligned northeast–southwest and are reflected in the pattern of major drainage and relief. The first-order structure is dominated by an imbricate system of NW-dipping thrust sheets involving sedimentary terrains of Palaeozoic and Mesozoic age (Figs. 2, 3), flanked in high ranges to the northwest and southeast by belts of discontinuous granitoids and associated metamorphites. The more prominent of these thrusts, the Reshūn Fault (Fig. 3) (Calkins et al., 1981; Pudsey et al., 1985), can be followed over several hundreds of kilometres. The pattern of relative tectonic transport throughout this thrust system is northwest-over-southeast. In some thrust sheets the presence of tectonically juxtaposed sedimentary successions of similar or overlapping geological ages but of markedly contrasted relative stratigraphic thickness and gross facies character, testifies to the large-scale lateral shortening that the region has evidently accommodated (Fig. 3).

The absence of cover rocks younger than mid-Cretaceous, impedes dating formation of the imbricate thrust-sheet system on stratigraphic grounds. Regional structural observations (e.g., Searle, 1991, pp. 100, 313) indicate that large-scale thrust processes did not occur prior to collision of India with Asia. A post-collisional origin of the imbricate system is further substantiated by our palaeomagnetic evidence for the occurrence of large-scale tilting and regional rotation during the latest Cainozoic.

6. Sampling

The palaeomagnetic results reported here stem from an extensive collection of block samples (333, Table 1) of upper Palaeozoic and Mesozoic rocks taken from five separate lithostratigraphic units, and three first-order thrust masses that

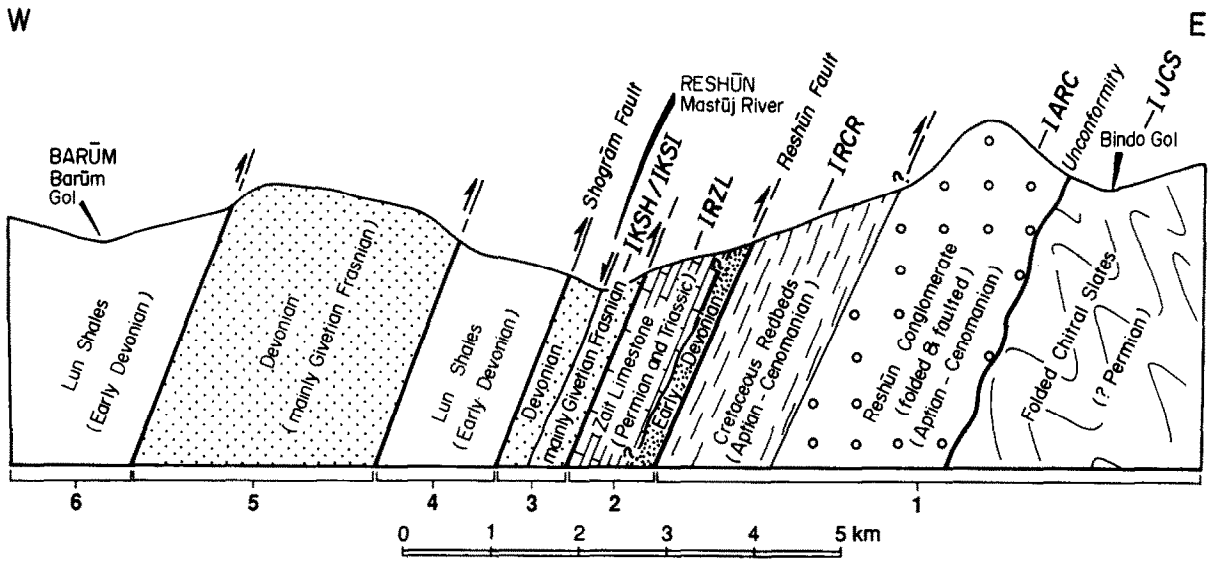
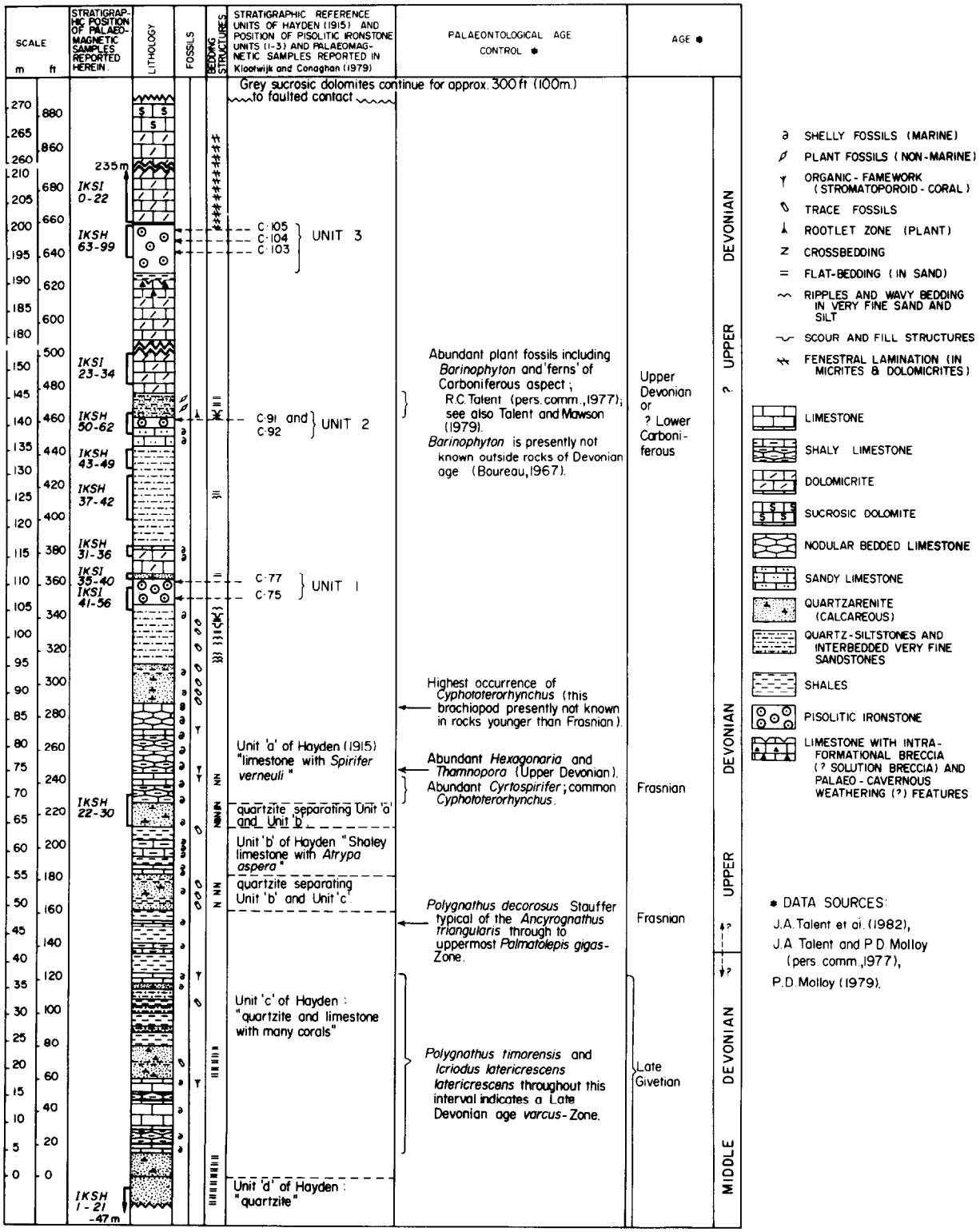


Fig. 3. Schematic cross section (oblique to strike) of the thrust pile in the Reshūn area, after Talent et al. (1982, plate 7). Sample acronyms (e.g. *IRZL*) are indicated with respect to the sampled thrust sheets (differentiated numerically along the base of the diagram).

Table 1
Sampling details

Stratigraphic unit and sample prefix (cf. Figs. 2B, 3 and 4)	Age (cf. Talent et al., 1982, and Fig. 3)	Sampling locality (cf. Fig. 2B)	Lithologies sampled	Stratigraphic thickness of sampled sections (m)	No. of field samples; (No. of lab. specimens)	Initial intensity (10^{-3} mA m^{-1})	Bedding attitude (S_0) dip (°), dip azimuth (°)
Reshūn Conglomerate (*IARC*)	mid-Cretaceous (Aptian–Cenomanian)	Awi, on right bank of Mastūj River opposite the village (72°21′E, 36°17′N)	Sandy mudstone/ siltstone, and lithic sandstone	5	32; (80)	241–1867	20–24/320–340
Redbeds (*IRCR*)	mid-Cretaceous (Aptian–Cenomanian)	Reshūn Gol; sampled at two sites on right bank, (72°7′E, 36°9′N)	Red mudstone	site 1: 16 site 2: 3	21; (40) 16; (37)	125–10,736	38.5/278 36/200 (= S_1?)
Zait Limestone (*IRZL*)	Permian through Early Triassic (?+)	Torman Gol, right bank (72°7′E, 36°10′N)	Dolomicrite	100	75; (80)	10–907	33.5/320–335
Chitral Slate (*IJCS*) [a]	Uncertain (?Permian)	Jumshili; road section on right bank of Jumshili Gol above village (72°4′E, 36°6′N)	Sandy slate, quartzite, sandstone	85	34; (73)	283–8874	21.5–74/143–215
Kurāgh Spur section (Fig. 4): Hayden's (1915) unit "d" (*IKSH*) and stratigraphically overlying succession (*IKSH, IKSI*)	Middle and Late Devonian (late Givetian–Frasnian and possibly Famennian)	Kurāgh Spur (72°9′E, 36°12′N)	Mid. Dev. (unit "d") quartzite; U.Dev: calcareous quartzose sandstone, siltstone, shaly limestone, sedimentary ironstones (some with pisolitic phases), dolomicrite	282	156; (349)	11–214,500	58–74/320–330

[a] Other structural details: conspicuously folded regionally by two phases of Folds (F_1, F_2); with associated slaty and crenulation cleavage; direction of stratigraphic facing generally difficult to determine, but is in some places reversed.
F_1 folds at sampling site have average plunge of 38°/032° on axial plane 85°/306°; F_2 folds not observed there.


SCALE
m
ft
STRATIGRAPHIC POSITION OF PALAEOMAGNETIC SAMPLES REPORTED HEREIN.
LITHOLOGY
FOSSILS
BEDDING STRUCTURES
STRATIGRAPHIC REFERENCE. UNITS OF HAYDEN (1915) AND POSITION OF PISOLITIC IRONSTONE UNITS (1-3) AND PALAEOMAGNETIC SAMPLES REPORTED IN Klootwijk and Conaghan (1979)
PALAEONTOLOGICAL AGE CONTROL *
AGE *
Grey sucrosic dolomites continue for approx. 300 ft (100m.) to faulted contact
235m
IKSI 0-22
IKSH 63-99
C·105
C·104
C·103
UNIT 3
IKSI 23-34
IKSH 50-62
C·91 and C·92
UNIT 2
IKSH 43-49
IKSH 37-42
IKSH 31-36
IKSI 35-40
IKSI 41-56
C·77
C·75
UNIT 1
IKSH 22-30
IKSH 1-21
-47m
Abundant plant fossils including Barinophyton and 'ferns' of Carboniferous aspect; R.C. Talent (pers. comm., 1977); see also Talent and Mawson (1979).
Barinophyton is presently not known outside rocks of Devonian age (Boureau, 1967).
Upper Devonian or ? Lower Carboniferous
Highest occurrence of Cyphoterorhynchus (this brachiopod presently not known in rocks younger than Frasnian).
Unit 'a' of Hayden (1915) "limestone with Spirifer verneuli"
Abundant Hexagonaria and Thamnopora (Upper Devonian).
Abundant Cyrtospirifer; common Cyphoterorhynchus.
Frasnian
quartzite separating Unit 'a' and Unit 'b'
Unit 'b' of Hayden "Shaley limestone with Atrypa aspera"
quartzite separating Unit 'b' and Unit 'c'
Polygnathus decorosus Stauffer typical of the Ancyrognathus triangularis through to uppermost Palmatolepis gigas-Zone.
Frasnian
Unit 'c' of Hayden: "quartzite and limestone with many corals"
Polygnathus timorensis and Icriodus latericrescens latericrescens throughout this interval indicates a Late Devonian age varcus-Zone.
Late Givetian
Unit 'd' of Hayden: "quartzite"
UPPER DEVONIAN
UPPER DEVONIAN
MIDDLE DEVONIAN
SHELLY FOSSILS (MARINE)
PLANT FOSSILS (NON-MARINE)
ORGANIC-FAMEWORK (STROMATOPOROID-CORAL)
TRACE FOSSILS
ROOTLET ZONE (PLANT)
CROSSBEDDING
FLAT-BEDDING (IN SAND)
RIPPLES AND WAVY BEDDING IN VERY FINE SAND AND SILT
SCOUR AND FILL STRUCTURES
FENESTRAL LAMINATION (IN MICRITES & DOLOMICRITES)
LIMESTONE
SHALY LIMESTONE
DOLOMICRITE
SUCROSIC DOLOMITE
NODULAR BEDDED LIMESTONE
SANDY LIMESTONE
QUARTZARENITE (CALCAREOUS)
QUARTZ-SILTSTONES AND INTERBEDDED VERY FINE SANDSTONES
SHALES
PISOLITIC IRONSTONE
LIMESTONE WITH INTRA-FORMATIONAL BRECCIA (? SOLUTION BRECCIA) AND PALAEO-CAVERNOUS WEATHERING (?) FEATURES
* DATA SOURCES:
J.A. Talent et al. (1982),
J.A. Talent and P.D. Molloy (pers. comm., 1977),
P.D. Molloy (1979).

juxtapose rocks of contrasting facies and stratigraphic age (thrust masses 1–3 of Fig. 3). The samples were oriented with a Brunton compass, with distance-bearing checks on possible local magnetic deflection. Salient details of these stratigraphic units, together with sampling information are summarized in Table 1.

Samples were taken from the left bank of the Mastūj River near the villages of Reshūn and Kurāgh and also from its right bank opposite Awi (Fig. 2B), with sampling of individual formations generally confined to separate thrust sheets. The study was directed in particular to extensive resampling of the Middle–Upper Devonian (Givetian to Frasnian–?Famennian) section exposed in the ridge section of "Kurāgh Spur" (Figs. 2B, 4). This section has been described in some detail in our earlier study on the chamositic ironstones (Klootwijk and Conaghan, 1979), and a downwards-extended stratigraphic log is reproduced here (Fig. 4).

A total of 156 block samples was taken over a stratigraphic interval of 282 m from the upper part of Hayden's (1915) quartzite unit "d" (Fig. 4, late Givetian) up to a massive dolomicritic section of Frasnian–?Famennian age which caps the Kurāgh Spur section. A wide variety of lithologies was sampled (Fig. 4), ranging from quartzite, sandstone and siltstone, to sandy limestone and dolomicrite. Our efforts were directed in particular to a detailed sampling of the three pisolitic ironstone horizons in the middle and upper parts of the section (16, 13 and 37 samples, respectively) on which we reported in our earlier paper (Klootwijk and Conaghan, 1979).

The Devonian thrust sheet sampled at Kurāgh Spur is tectonically underlain by a belt of massive carbonates (principally dolomites, "Zait Limestone", Fig. 3) cropping out in a high cliff on the left bank of the Mastūj River between Reshūn and Kurāgh (Fig. 2B). This approximately 600-m-thick section of Zait Limestone of Permian–Triassic age overlies Lower Devonian sandstones with an apparently sedimentary contact (Talent et al., 1982). A total of 75 block samples was taken from the Zait Limestone through 100 m of stratigraphic section in cliffs on the right bank of Torman Gol (Fig. 2B) in locally well-bedded to massive and heavily jointed blue-grey carbonates forming the base of the section.

The next underlying thrust sheet (Fig. 3) comprises an approximately 1-km-thick unit of bright-red mudstone, exposed along the left bank of the Mastūj River and in the north wall of Reshūn Gol (Fig. 2B). *Orbitulina*-bearing shale and argillaceous limestone near the top of the unit give an Aptian–Cenomanian age (Talent et al., 1982; and P.G. Quilty, written commun. to P.J.C., 1978). The red mudstone is very uniform in appearance: bedding could be determined only locally on the basis of sporadically present, coarser-grained intercalations. The mudstone has a complex internal structure and stratigraphic facing could not be established. A total of 37 block samples was taken at two sites on the right bank of Reshūn Gol, about 200 m apart stratigraphically, covering 16 m (site 1) and 3 m (site 2) of section, respectively. Bedding at site 1 was established from a coarser-grained intercalation; no bedding markers were identified at site 2 and it is possible that the conspicuous planar structure at this site represents cleavage and not bedding.

The redbed unit is tectonically, and probably also stratigraphically, underlain by the massive Reshūn Conglomerate of Aptian–Cenomanian age (age attribution by P.G. Quilty, written commun. to P.J.C., 1978). Near Awi (Fig. 2B), 32 block samples were taken through 5 m of stratigraphic section from two siltstone-sandstone intercalations within the Reshun Conglomerate, several hundred metres below the thrust contact with the redbeds.

The structurally lowermost unit sampled, the Chitral Slate Complex, is a typical quartz flysh of uncertain age; age attributions range from pre-Devonian (Precambrian–early Palaeozoic) (Ta-

Fig. 4. Stratigraphic log of the Middle–Upper Devonian part of the succession sampled at Kurāgh Spur. Sampled intervals are indicated and stratigraphic thicknesses are referred to the top of Hayden's (1915) quartzite unit "d". Section logged by Conaghan, August and September 1975.

hirkheli, 1982) through Permian (Talent et al., 1982), to Jurassic (Pudsey et al., 1985) and Cretaceous (Calkins et al., 1981). It is characterized by conspicuous isoclinal folds and underlies the Reshūn Conglomerate with marked erosional unconformity (Fig. 3). Thirty-four block samples were taken over 85 m of stratigraphic section from sandy shale and flaggy quartzitic sandstone along the roadside near Jumshili (Fig. 2B). Regional folding in the Chitral Slate plunges moderately towards north-northeast (Table 1). Facing of the beds at the Jumshili sampling site could not be established, and was assumed upward in the tilt corrections applied.

7. Analysis and results

Cores were drilled from the 333 block samples at the Quetta offices of the Geological Survey of Pakistan. At the Black Mountain palaeomagnetic laboratory in Canberra, the cores were rinsed in diluted HCl to remove any metal contaminants that may have adheared during coring. Three or more specimens, 2.5 cm in diameter and 2.2 cm height, were sliced from the cores of each block sample. All specimens were measured for initial natural remanent magnetization (NRM) and bulk susceptibility, and on this basis a total of 659 specimens was selected for demagnetization studies.

Thermal demagnetization and analysis of results were carried out according to procedures described in some detail elsewhere (Klootwijk and Bingham, 1980; Klootwijk et al., 1981). Progressive thermal demagnetization was carried out in large-volume furnaces (McElhinny et al., 1971) with a feedback-controlled ten-coil Helmholtz-Rubens system maintaining the ambient magnetic field in the heating chamber to less than 10 nT during the heating–cooling cycle (about 2 h). All NRM measurements were made on a two-axis ScT cryogenic magnetometer interfaced with a HP-2100 minicomputer. Pilot specimens, generally one for every second or third consecutive sample throughout a profile, were progressively

Table 2
Summary of palaeomagnetic directions and pole positions
a. Mean specimen direction [a]

Formation	In geographic coordinates				In stratigraphic coordinates				N [b]	Polarity (N/R) [c]
	Decl. (°)	Incl. (°)	k	α_{95} (°)	Decl. (°)	Incl. (°)	k	α_{95} (°)		
A: overprint component (late Cainozoic)										
Reshūn Conglomerate (*IARC*)	347	+53.5	21	5.5	–	–	–	–	34	N
Redbeds (*IRCR*), sites 1 and 2	338.5	+64	14.5	6.5	–	–	–	–	38	N
Chitral Slate (*IJCS*)	333.5	+62	10	13	–	–	–	–	14	N
Chitral Slate [d] (*IJCS*)	158.5	+14.5	2.5	31.5	140	−55	9.5	13.5	14	2/12
Zait Limestone (*IRZL*)	3	+54	33.5	4.5	–	–	–	–	31	N
Kurāgh Spur section [e] (*IKSH,IKSI*)	356.6	+51	39.5	2	–	–	–	–	146	124/22
Kurāgh Spur section [f] (*IKSH,IKSI*)	0	+53	31.5	3.5	–	–	–	–	61	N
B: Secondary component (early Tertiary)										
Reshūn Conglomerate	306	+21.0	6	10	308	+3.5	6	9.5	44	41/3
Redbeds, site 1: upward facing [g]	274	+43.5	4.5	17	272.5	+4	7	13	21	N
Redbeds, site 1: downward facing [g]	274	+43.5	4.5	17	103.5	−4	7	13	21	R
Chitral Slate [d]	138	+61	6	20.5	139	−9	8.5	16.5	11	R
Zait Limestone	134.5	−39	10.5	7.5	136.5	−2.5	10.5	7.5	40	17/13
Kurāgh Spur section	298.5	+68	16.5	2.5	312.5	+9	17	2.5	232	171/61
Mean of rows 7, 9–11 (see Table 2b)	–	–	–	–	314.1	+6	198.2	6.5	4 [i]	–
C: Components of uncertain origin										
Redbeds, site 1: upward facing [g]	330.5	−29.5	12	8	359	−45.5	12	8	30	N
Redbeds, site 1: downward facing [g]	330.5	−29.5	12	8	17	+45.5	12	8	30	R
Redbeds, site 2 [h]	67	+76.5	14	7	176.5	+61.5	13.5	7	33	R

Table 2 (continued)
b. S-pole position [a]

Formation	In geographic coordinates				In stratigraphic coordinates				Palaeo-latitude (°)	Pole no. in Fig. 9
	Long. (°E)	Lat. (°S)	*dp* (°)	*dm* (°)	Long (°E)	Lat. (°S)	*dp* (°)	*dm* (°)		
A: overprint component (late Cainozoic)										
Reshūn Conglomerate (*IARC*)	155.5	79	5.5	7.5	–	–	–	–	–	1
Redbeds (*IRCR*), sites 1 and 2	200	71	8	10	–	–	–	–	–	2
Chitral Slate (*IJCS*)	190	68.5	15.5	20	–	–	–	–	–	3
Chitral Slate [d] (*IJCS*)	–	–	–	–	174.5	57.5	13.5	19	35.5N	3′
Zait Limestone (*IRZL*)	18	87	4.5	6.5	–	–	–	–	–	4
Kurāgh Spur section [e] (*IKSH,IKSI*)	108	84.5	1.5	2.5	–	–	–	–	–	5
Kurāgh Spur section [f] (*IKSH,IKSI*)	70.5	87.5	3	4.5	–	–	–	–	–	6
B: Secondary component (early Tertiary)										
Reshūn Conglomerate	–	–	–	–	138.5	31.5	5	9.5	1.5N	7
Redbeds, site 1: upward facing [g]	–	–	–	–	162.5	3	6.5	13.5	2N	8′
Redbeds, site 1: downward facing [g]	–	–	–	–	155.5	12	6.5	13	2N	8
Chitral Slate [d]	–	–	–	–	131.5	41	8.5	16.5	4.5N	9
Zait Limestone	–	–	–	–	131	36.5	3.5	7.5	1N	10
Kurāgh Spur section	–	–	–	–	138	36	1	2.5	4.5N	11
Mean of poles in rows 7, 9–11	–	–	–	–	135.3	36.1	3.3	6.5	3.0N	12
C: Components of uncertain origin										
Redbeds, site 1: upward facing [g]	–	–	–	–	73	27	6.5	10	27S	13′
Redbeds, site 1: downward facing [g]	–	–	–	–	10	73	6.5	10	27N	13 [j]
Redbeds, site 2 [h]	–	–	–	–	75	11	8.5	11	43S	14 [j]

[a] Decl. = declination; Incl. = inclination; k = estimate of parameter; α_{95} = half-angle of the circle of 95% confidence about the mean direction; Long. = longitude; Lat. = latitude; *dp*, *dm* = half-angles of the oval of 95% confidence about the pole position.
[b] Number of specimens used in the analysis (except where indicated otherwise).
[c] N/R: number of specimens with normal/reverse polarity component.
[d] Corrected for F_1 folds with plunging fold axis (see Table 1).
[e] Component with low blocking temperature range (below 300°C).
[f] Component with high blocking temperature range (above 600°C).
[g] Arbitary options due to lack of stratigraphic facing control (see text).
[h] Probably corrected with reference to slaty cleavage rather than bedding.
[i] Number of thrust sheets.
[j] Not depicted in Fig. 9 (see text).

demagnetized in 14–21 steps from room temperature up to 600 or 695°C or until the NRM intensity had dropped too low ($< 2 \times 10^{-3}$ mA m^{-1}) for reproducable measurements on the ScT cryogenic magnetometer. Remaining specimens were partially progressively demagnetized in 12–15 steps over optimal temperature intervals depending on their blocking temperature spectra. Bulk susceptibility was measured on a Digico bridge after each heating–NRM-measurement step to monitor magnetochemical change (Fig. 5). The carbonate specimens in particular showed excessive susceptibility increases after heating above 400–500°C, which often impeded further remanence measurement. Individual magnetic components were identified visually from Zijderveld (1967) plots and their directions were determined with an adapted version of Kirschvink's (1980) principal component analysis program.

Intensities of initial NRM varied widely between 10^{-2} and 2×10^{2} mA m^{-1}, reflecting the wide range of lithologies studied. The dolomicrites from the Upper Devonian section at Kurāgh Spur and from the Zait Limestone had a very low magnetic intensity resulting in some cases in premature abandonment of a progressive demagnetization cycle. Consistent within-sample and within-section results were obtained from, in par-

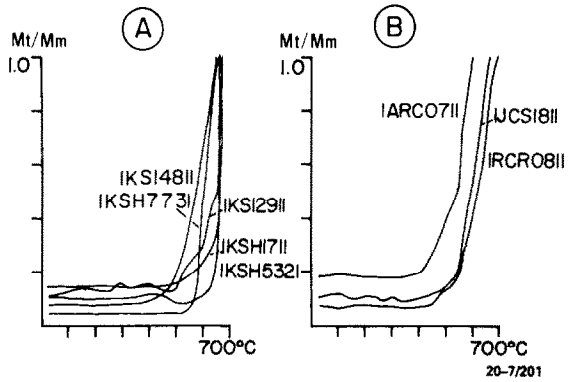


Fig. 5. Normalized curves showing changes in magnetic susceptibility during progressive thermal demagnetization for some representative specimens. Individual data points omitted for clarity.

ticular, the Middle–Upper Devonian section at Kurāgh Spur and the Cretaceous redbeds. Results from a considerable number of specimens from the other formations had to be rejected because of inconsistent within-sample results (Chitral Slate Complex and to a lesser extent the Reshūn Conglomerate), or because their NRM intensity had dropped below measurable levels (Zait Limestone) before individual components could be determined.

Three components of magnetization could be identified; two of them (A and B) were present throughout all formations studied (Table 2; Fig. 8).

7.1. Component A

This is a recent field component, with a normal polarity throughout most of the formations, but with a reverse polarity in part of the Kurāgh Spur section and in the Chitral Slate Complex. In the latter formation, this component is of recent, but pre-tilt, origin. The intensity of this recent field component generally was 30–40% of initial NRM, but represented up to 70% in the dolomicrites from the Zait Limestone and from the Kurāgh Spur section (Figs. 6, 7A–F, H–L, N–P; note that the data shown in the Zijderveld diagrams are in geographical coordinates, i.e., the data are not corrected for tilting). The recent field component was generally removed below 200–300°C. Part of the Kurāgh Spur section, however, showed another harder recent field component which was broken down above 600°C (Table 2; Figs. 6, 7B, D, E, H, 8A).

7.2. Component B

This is a characteristic component of both normal (Figs. 7A, B, E–I, K–O, R, S) and reverse polarity (Fig. 7C–H, J, K) and with northwestward- and southeastward-pointing tilt-corrected

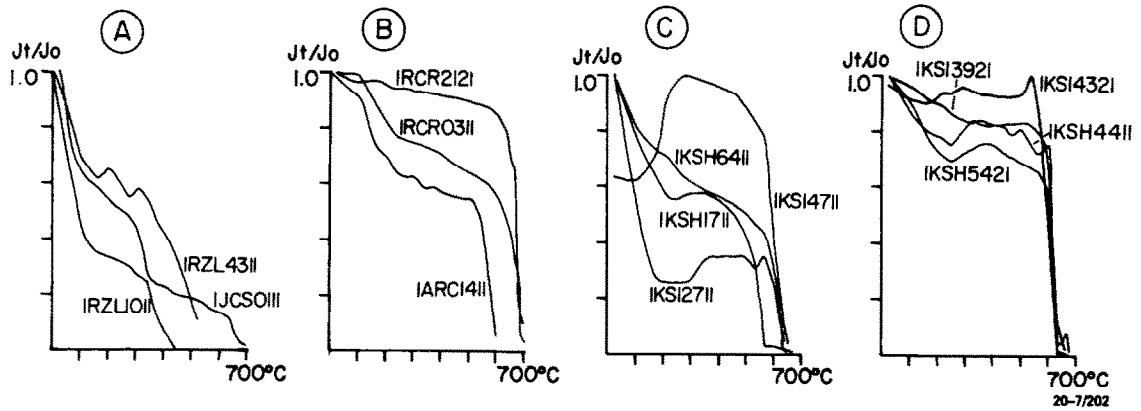


Fig. 6. Normalized curves showing decay of natural remanent magnetization intensity during progressive thermal demagnetization for some representative specimens. Individual data points omitted for clarity.

directions with very shallow inclinations (Table 2; Fig. 8). This component was broken down in general, between 300 and 600°C (Figs. 6, 7A–H, L–O, R, S, 8B), and between 300 and 500°C in the Zait Limestone (Figs. 6, 7I–K, 8B).

7.3. Component C

This third component is dominantly present in and confined to the red mudstone, with good within-site but different between-site groupings,

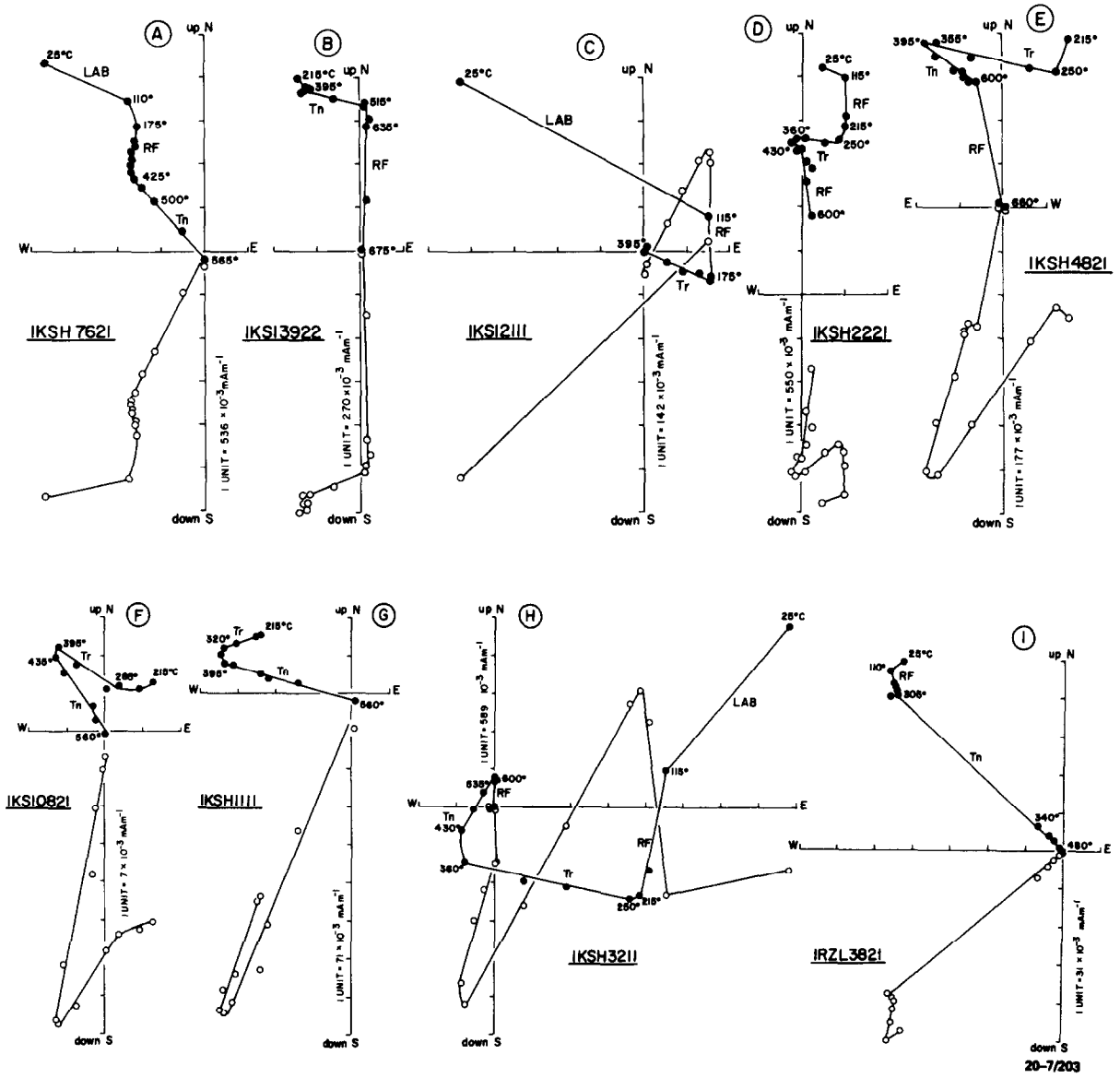


Fig. 7. Zijderveld (1967) diagrams of some representative specimens in geographic coordinates (not corrected for tilting). The points indicate successive positions—in orthogonal projection—of the end points of the resultant magnetization vector during progressive thermal demagnetization. ○ indicates projections on the vertical-E–W plane, ● indicates projections on the horizontal plane. Successive demagnetization temperatures indicated in degrees Celsius. *LAB* = laboratory induced component; *RF* = recent field component; *T* = collision-related secondary component; *P* = component of unclear origin in the red mudstones (see text, component C); *CO* = composite breakdown of two components; $_n$ = normal polarity; $_r$ = reverse polarity.

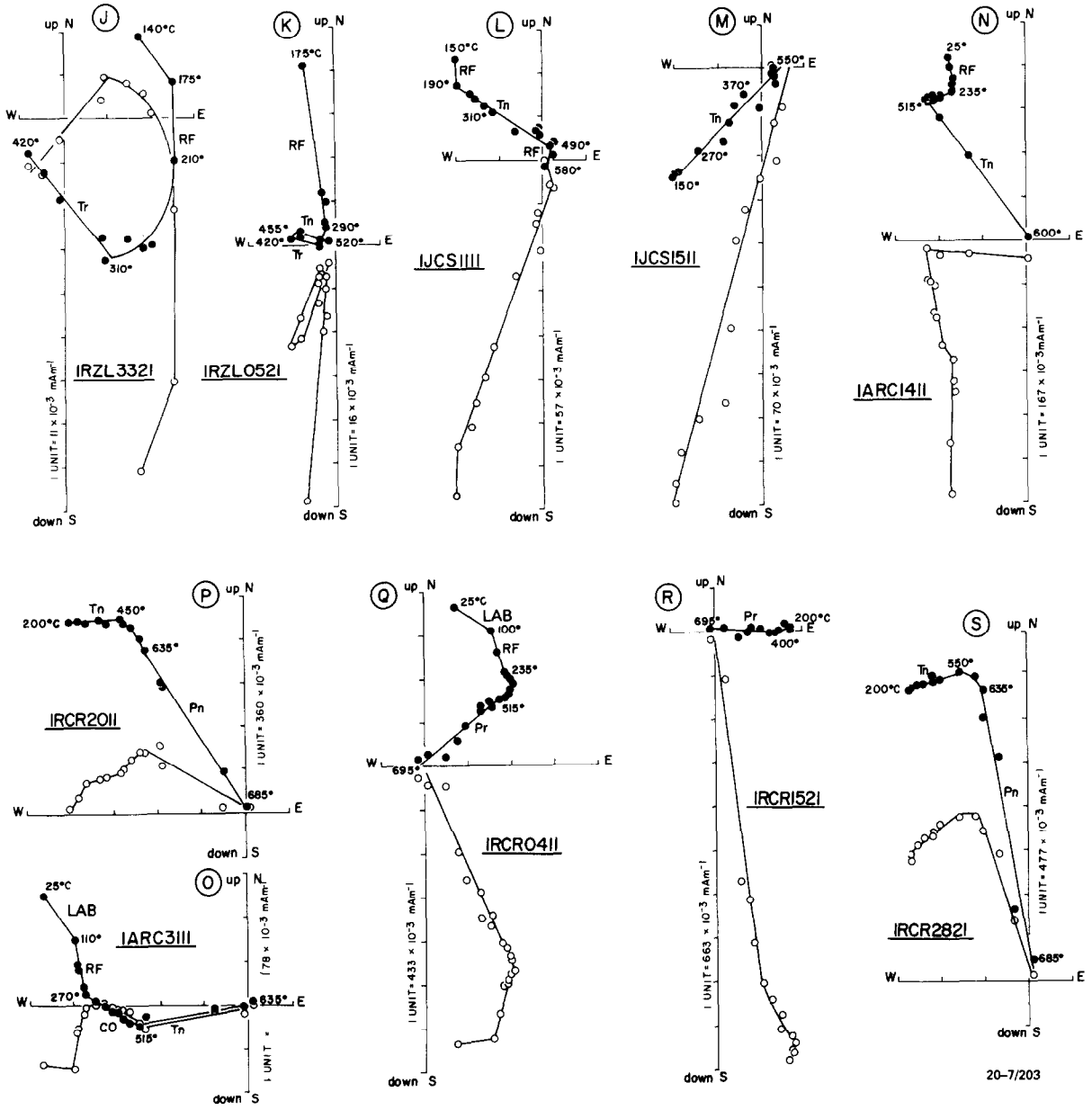


Fig. 7 (continued).

and is broken down between 600 and 670°C (Figs. 6, 7P–S).

8. Interpretation

The pole positions for the characteristic component (B; note stratigraphic coordinates, Table 2) in the Reshūn Conglomerate, the Chitral Slate Complex, the Zait Limestone and the Middle–Upper Devonian succession at Kurāgh Spur, are all close to the previously determined (Klootwijk and Conaghan, 1979) preliminary pole position for the pisolitic ironstones of the Kurāgh Spur succession (Table 2; Fig. 9). Pole positions for the B component in the red mudstone (site 1: alternative pole positions 8′ and 8 are indicated, re-

spectively, for upward- and downward-facing attitudes of the beds in Table 2 and Fig. 9) are offset to the east of this main grouping. All B-component pole positions, however, fall within a swath, of secondary pole positions, defined by previous studies (Klootwijk et al., 1979, 1986) and complemented recently (Klootwijk et al., 1991) from regions both north and south of the Indus–Tsangpo Suture zone (Figs. 9, 10), that are attributed to secondary magnetization components acquired upon initial India–Asia contact. This and conodont CAI evidence for heating above 300°C, lead us to also interpret a contact-related, pre-tilting, origin for the B-component magnetization. Such an overprint origin is further supported by observations in the sampled area of partially reset zircon fission-track ages around 68–55 Ma (Zeitler, 1985, fig. 5). Presence of both normal and reverse polarity components indicates acquisition of this secondary component over a prolonged timespan. Alternative interpretation of a primary origin would imply absence of significant latitudinal or rotational movement of these rocks from Middle Devonian to mid-Cretaceous, but such movements would be expected for either a Gondwanan or "Laurasian" affinity of Chitral (e.g, Besse and Courtillot, 1991; Van der Voo, 1993). We think it unlikely that this secondary magnetization results from internal deformation processes related to thrust-sheet translation (e.g., Hirt et al., 1986; McWhinnie et al., 1990) such as grain rotation and formation of stretching lineations. The effectiveness of such processes to deflect the (primary) magnetization direction strongly depends upon competency of the rocks and intensity of the deformation process. Absence of penetrative deformation in at least part of the succession points to mild deformation conditions, and competency of the rocks varies considerably across the wide range of lithologies and ages studied (Table 1). Hence, dispersed rather than grouped (re)magnetization directions would be expected. Yet we observe close directional agreement for the B-component remagnetization throughout the thrust pile (Table 2; Figs. 8, 9). Consequently, we regard remagnetization through thermo-chemical and/or thermo-viscous remagnetization processes more likely, as conodont CAI studies (see above) and fission-track cooling-age studies (Zeitler, 1985) indicate past temperatures to have been in excess of 300°C.

The offset positions of the B-component poles (Fig. 9) with respect to the Indian apparent polar wander path (APWP) are interpreted to reflect

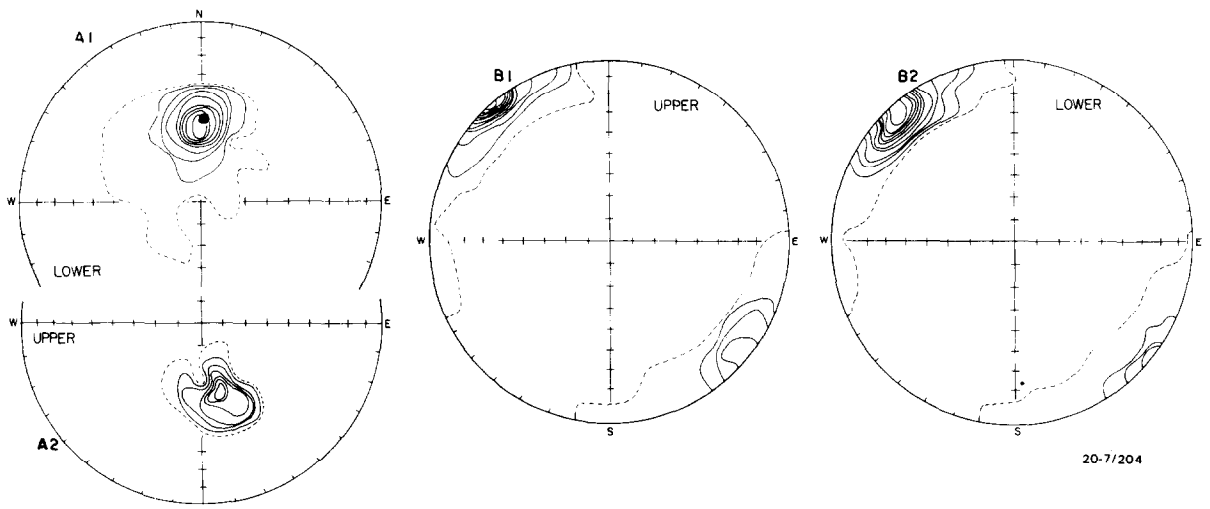


Fig. 8. Frequency distribution of magnetization directions combined for all formations studied, counted with a 1%-circle. A = recent field components in geographic coordinates. B = collision-related overprint components in stratigraphic coordinates (i.e., corrected for bedding). Frequency intervals: $A1 = 10$; $A2 = 2$; $B1 = 5$; $B2 = 5$. • in $A1$ indicates the local dipole field direction. Equal-area projection.

rotations of the sampled thrust sheets with respect to the Indian subcontinent that occurred during India's post-collisional convergence with southern Asia. The APWP shown in Fig. 9 was recently determined (Klootwijk et al., 1991, 1992) from an extensive set of palaeolatitude data from the Ninetyeast Ridge (ODP Leg 121) that reflect the northward movement of the Indian plate dur-

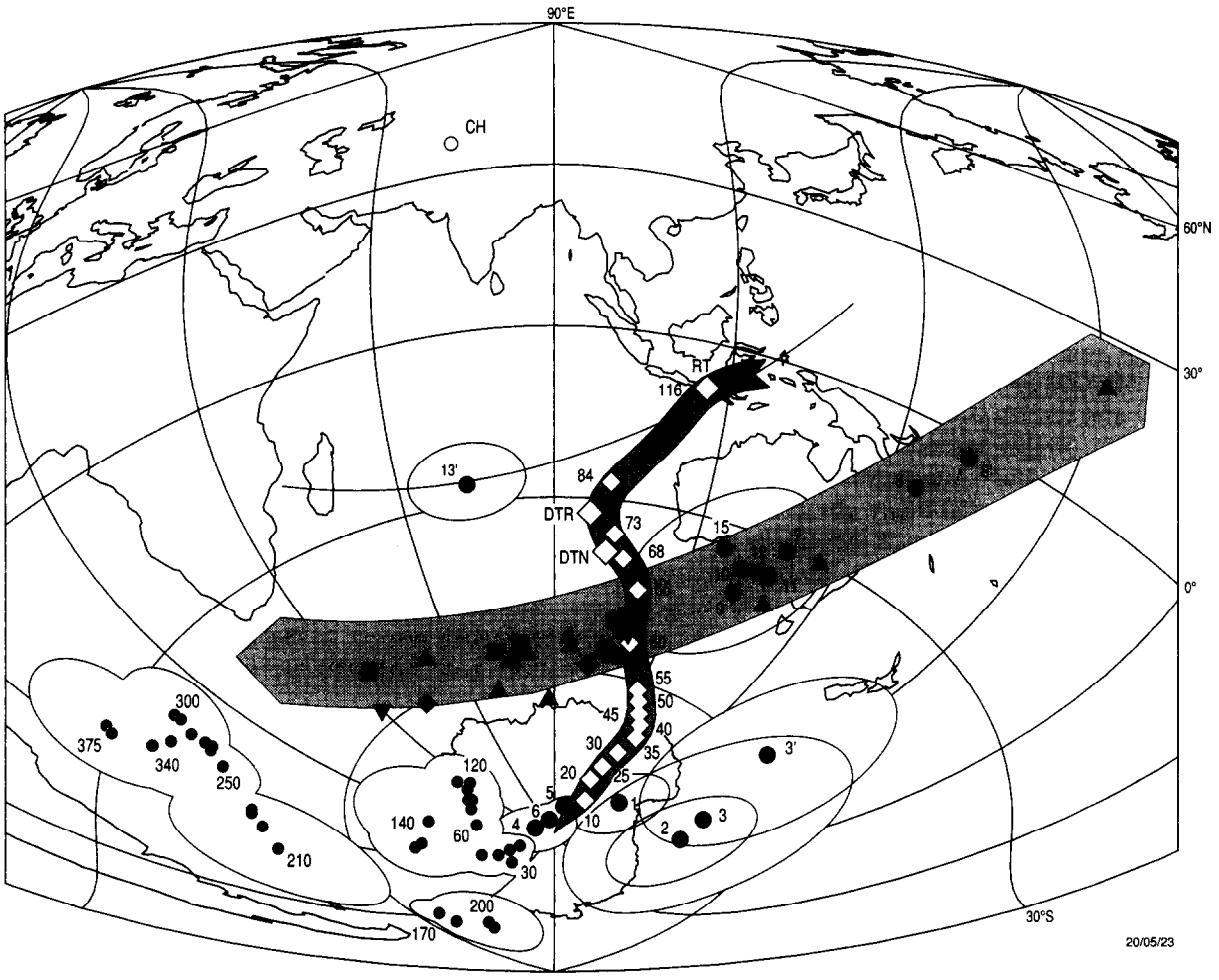


Fig. 9. Comparison of pole positions obtained in this study (numbered large dots, sampled region tagged *CH*) for the recent field components (1–6; 3′ indicates a reverse polarity component, see Table 2) and the collision-related overprints (7–11; 8 and 8′ indicate two alternative pole positions due to uncertain facing, see Table 2), and the mean of poles 7 and 9–11 (12, cf. Table 2) with: (i) the Late Cretaceous–Cainozoic segment of the Indian APWP (small diamonds with dates indicated) according to Klootwijk (1984) and Klootwijk et al. (1991; 1992, table 2), and pole positions (large diamonds) for the Deccan Traps (*DTN*, *DTR*: 70–65 Ma) and the Rajmahal Traps (*RT*: 117–116 Ma); (ii) a swath of collision-related overprints, see Klootwijk et al. (1991, table 9) observed in the Ladakh Intrusives, the Indus Molasse and the Dras Flyschoids of the Ladakh region (▲), the Tibetan Sedimentary Series and the Nawakot Complex of central Nepal (■), the Dingri Limestone of southern Tibet and Jurassic limestone near Lhasa (♦), and the Moghul Kot Formation, the Nowshera Reef and the Nammal Gorge section (Haag and Heller, 1991) of Pakistan (▼); and (iii) the Eurasian APWP shown as stippled fields and dated pole positions (●) at bottom-left [Besse and Courtillot, 1991 (200–0 Ma); Irving and Irving, 1982 (330–210 Ma); Irving, 1977 (375–340 Ma)]. Pole *13′* represents a hard magnetic component in the Redbeds site 1 of uncertain interpretation (see text and Table 2). Pole *15* is the preliminary pole position documented by Klootwijk and Conaghan (1979). The swath formed by the loci of the collision-related secondary components intersects the Indian APWP around the KTB. Compare with Fig. 10.

ing the past 80 Myr, and from Eulerian rotation data that define India's movement with respect to a mantle-fixed hotspot frame. This swath of collision-related pole positions intersects the Indian APWP in the latest Cretaceous–Palaeocene interval. Three conclusions can be drawn from comparison of the Chitral data with this APWP:

(1) The characteristic components observed in the present study are of secondary origin and were acquired at about the time of the KTB. This magnetic overprinting is probably related to resetting of zircon fission tracks around 68–55 Ma (cf. Zeitler, 1985, fig. 5). Such an overprint origin revokes our earlier interpretation of a primary origin for the characteristic component, based on interpretation of preliminary results from the Devonian pisolitic ironstones from Kurāgh Spur (Klootwijk and Conaghan, 1979), and conse-

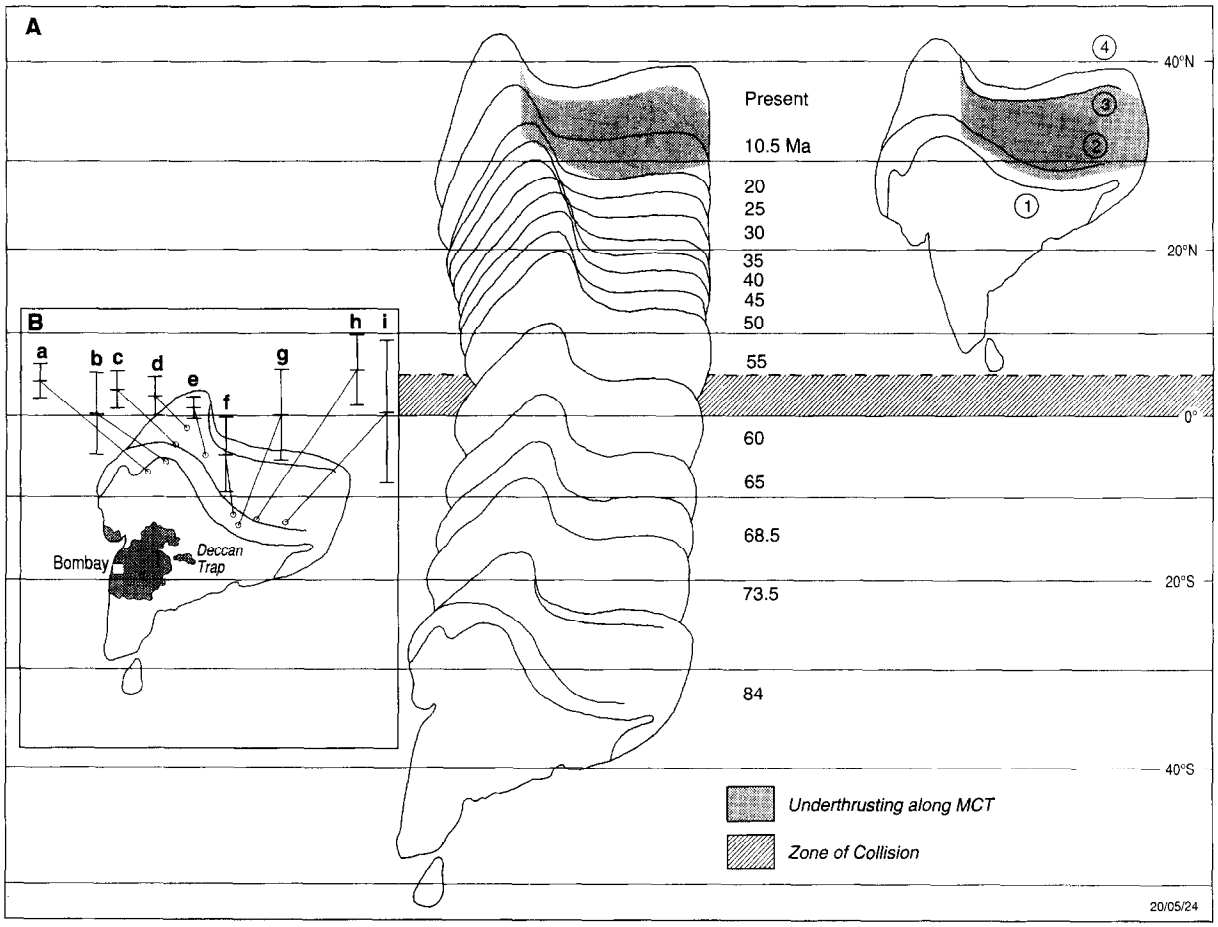
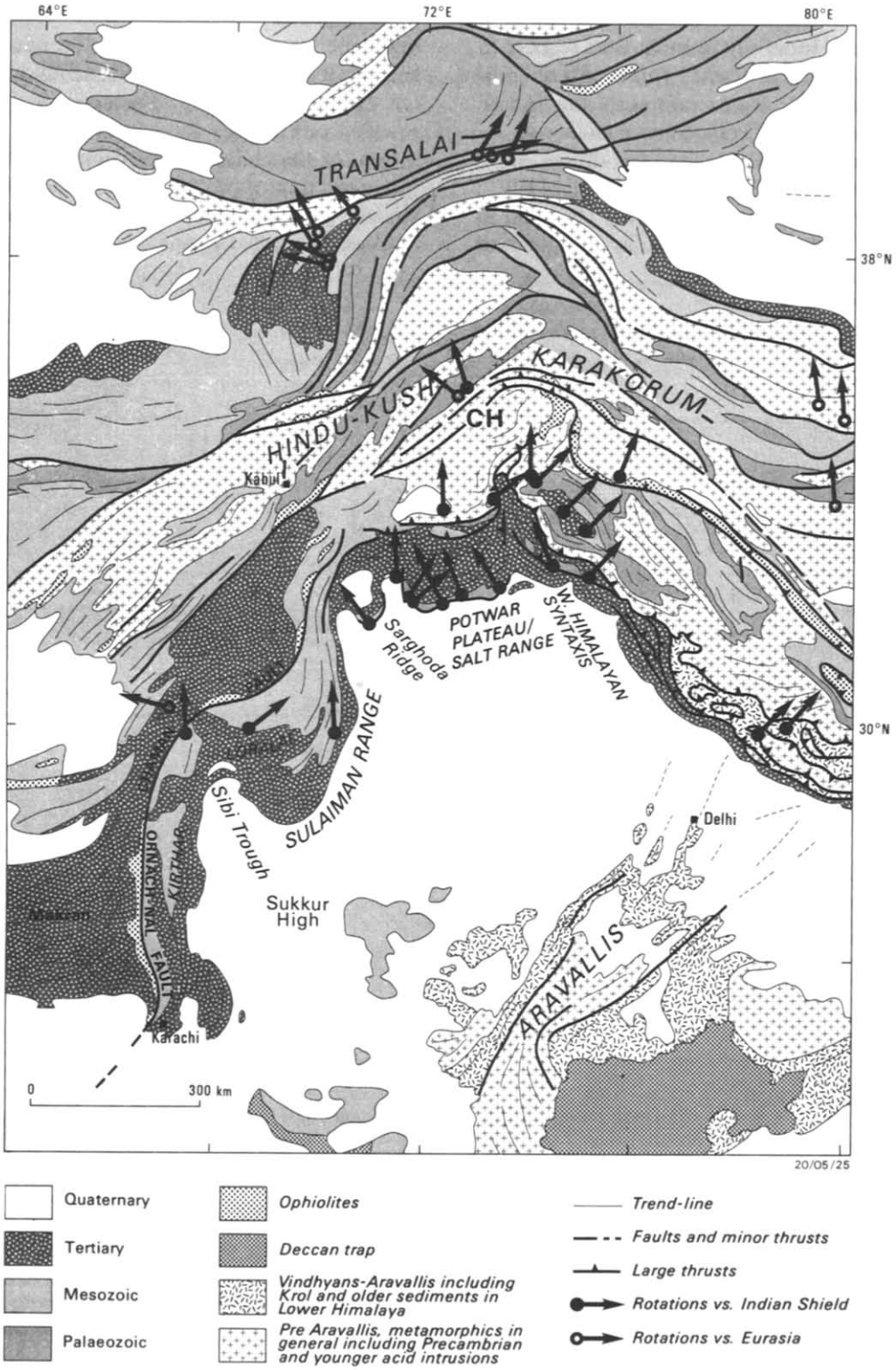


Fig. 10. (A) Northward movement of Greater India according to its Late Cretaceous–Cainozoic APWP (from Klootwijk et al., 1991, fig. 57, with additions). Outlines within Greater India represent from south to north: (*1*) the Himalayan mountain front; (*2*) the present-day position of the Indus–Tsangpo Suture rotated with the outline of Peninsular India; (*3*) reconstructed position of southern boundary of the Tethyan Himalayan region presently delineated by the Main Central Thrust (MCT); (*4*) minimal northern extent of Greater India as described in the text. The shaded equatorial-to-low-northern palaeolatitudinal belt is based on collision-related secondary magnetization components detailed in Figs. 9 and 10B. (B) Greater India's palaeoposition at 65 Ma and palaeolatitudes (with error bars at the 95% confidence level) defined by the collision-related secondary magnetization components (see Klootwijk et al. 1991, table 9, for references): *a* = Moghul Kot Formation; *b* = Nammal Gorge section (Haag and Heller, 1991); *c* = Nowshera Reef; *d* = Chitral data (herein); *e* = Indus Molasse–Dras Flyshoids; *f* = Tibetan sediments; *g* = Nawakot Complex; *h* = Dingri Limestone; *i* = Jurassic Limestone, Lhasa. Sample locations *a–i* are shown in their present position relative to the Indian Shield, not in their original position within Greater India.


64°E
72°E
80°E
38°N
30°N
TRANSALAI
HINDU-KUSH
KARAKORUM
CH
Kabul
Sarghoda Ridge
POTWAR PLATEAU/ SALT RANGE
W. HIMALAYAN SYNTAXIS
SULAIMAN RANGE
Sibi Trough
ORNACH-NAL FAULT
MAKRAN
Makran
Sukkur High
Delhi
ARAVALLIS
Karachi
0
300 km
20/05/25
Quaternary
Tertiary
Mesozoic
Palaeozoic
Ophiolites
Deccan trap
Vindhyans-Aravallis including Krol and older sediments in Lower Himalaya
Pre Aravallis, metamorphics in general including Precambrian and younger acid intrusions
Trend-line
Faults and minor thrusts
Large thrusts
Rotations vs. Indian Shield
Rotations vs. Eurasia

quently also revokes our earlier conclusion on the basis of those data of a Late Devonian "Laurasian" affinity of the Eastern Hindukush. The secondary origin of these components, now evident, does not allow distinction between a "Laurasian"-versus-Gondwanan affinity of the sampled region prior to the KTB, and cannot be used to test the alleged Gondwanan origin of the eastern sector of Stöcklin's "Central Domain" or Šengör's Cimmerian continent.

(2) The results provide further support for initial contact between northwestern Greater India and southern Asia being already established at, or before, the KTB. Klootwijk (1984) argued previously, on the basis of a comparison between then available pole positions for overprints attributed to initial India–Asia contact, mainly from Ladakh and central Nepal, and a version of India's Tertiary APWP then based on DSDP palaeolatitude data, that initial contact was established already by Late Palaeocene–Early Eocene times. In a subsequent synthesis Klootwijk et al. (1985) updated the timing of collision to initial contact prior to 62 Ma, based on comparison of the overprint magnetizations with a synthetic APWP derived from India's motion versus a mantle-fixed hotspot frame (Patriat, pers. commun., 1983; Patriat and Achache, 1984). Subsequently, Klootwijk et al. (1991, 1992) detailed new evidence that constrains initial contact to a time at, or before, the KTB—an interpretation also reached from faunal evidence (Jaeger et al., 1989). This conclusion is based on comparison of (a) the newly revised Indian APWP for the past 80 Myr (Fig. 9), with (b) a more extensive set of collision-attributed overprints (Figs. 9, 10), and with (c) a revised estimate of the northern extent of Greater India (Fig. 10).

The collision-related overprints have been observed in regions both north (Ladakh, Lhasa) and south of the Indus–Tsangpo Suture zone; and, in the present case of Chitral, also north of the Northern Kohistan Suture (Pudsey et al., 1985) which separates Stöcklin's "Axial Ophiolite Belt" from the "Central Domain". Greater India's northern extent is a minimal estimate based on evidence from balanced cross sections for crustal shortening in the Western Himalayan Syntaxis and on palaeomagnetic evidence for counterclockwise rotational underthrusting along the Main Central Thrust (Klootwijk et al., 1985, 1991). The new APWP indicates that at 65 Ma Greater India's northwestern margin had already entered equatorial-to-low-northern latitudes, as also indicated by the collision-related overprints. It can be concluded, therefore, that initial India–Asia contact was established already at, or before, the KTB. Further geological arguments in support of this palaeomagnetically based conclusion are discussed at some length elsewhere (Klootwijk et al., 1991, 1992). These include faunal evidence for an India–Asia connection during extrusion of the Deccan Traps at, or before, the KTB (mainly during Chron 29R or more likely Chron 31R), the geological and tectonic record of the Zanskar region of the northwestern Himalaya, collision-related rejuvenation of major fault zones (including the Narmada–Son Fault Zone leading to eruption of the Deccan Traps), and a fundamental reorganization of the sea-floor spreading pattern in the western Indian Ocean at about 65 Ma.

(3) The sampled thrust sheets have undergone counterclockwise rotations of 10–30° with respect to the Indian Shield and 60–70° with respect to Eurasia (Fig. 11). The collision-related pole positions for Chitral (Fig. 9) are east of the Indian APWP and show a much greater offset with respect to the Eurasian APWP (Besse and Courtillot, 1991). They indicate rotations that are surprisingly coherent for the individual thrust sheets, with the exception of the data from the incompetent Cretaceous red mudstones. Though only representative of the sedimentary superstructure, these rotations conform to a general pattern of counterclockwise rotations in the western limb of the Hindukush–Pamir–Karakorum syntaxial bend

Fig. 11. Overview of the magnitude and sense of rotation of thrust sheets, including those sampled in the present study, versus the Indian Shield (arrows with •, angle with respect to north) or Eurasia (arrows with ○, ibid) as based on comparison of palaeomagnetic data with the Indian and the Eurasian APWP's as detailed in Fig. 9 (from Klootwijk et al., 1986, fig. 9, with additions). *Ch* (= Chitral) indicates the studied region in the Eastern Hindukush.

and clockwise rotations in its eastern limb, clearly outlining the secondary origin of this arc (Klootwijk et al., 1985, 1986, 1991; Treloar et al., 1992). Comparable counterclockwise offsets with respect to the Indian APWP are shown also by pole positions for some of the recent field components (A) observed in the thrust sheets (Fig. 9). This indicates the young, late Cainozoic, ongoing nature of tectonic activity in the Chitral region.

Interpretation of the hard magnetization component (C) in the red mudstone is not clear for lack of stratigraphic facing control. No acceptable interpretation can be forwarded for the aberrant direction of this component in the site 2 samples (Table 2). Clear bedding markers have not been observed at this site and the observed fabric may be tectonically induced. Facing at site 1 could not be resolved from interpretation of the secondary component (B), with both the upward- and the downward-facing option resulting in acceptable interpretations for the pole positions (Table 2, Fig. 9, poles 8′ and 8, respectively). Interpretations for the hard characteristic component (C) at site 1 are likewise ambiguous. The upward-facing option invokes an Indian-like pole position (Table 2, Fig. 9, pole 13′) at a moderate southern palaeolatitude. This conflicts with geological arguments that the red mudstone of Aptian–Cenomanian age postdates the Neo-Cimmerian (i.e., Late Jurassic–Early Cretaceous according to Boulin, 1981, pp. 265–266) suturing of the "Central Domain" with southern Asia, then at low northern palaeolatitudes according to available palaeomagnetic data (Besse and Courtillot, 1991) (Fig. 9 herein). This option would also invoke a clockwise rotation rather than the counterclockwise rotations indicated by components B and A for the red mudstone and also for the more competent beds of the thrust pile. The downward-facing option would result in a pole position that can be interpreted either as a primary magnetization of Eurasian origin and absence of rotation, or can be interpreted as a recent field component with clockwise rotation of the tectonic unit (Table 2, pole 13). Both possibilities, likewise, conflict with the predominant evidence for counterclockwise rotation of the thrust pile.

9. Conclusions

Thermal demagnetization studies of an extensive sample collection from Middle Devonian to mid-Cretaceous sedimentary rocks throughout the thrust pile in Chitral in the Eastern Hindukush show two interpretable magnetization components: a recent field component of mainly post-tectonic origin (A), and a characteristic component of both normal and reverse polarity and with equatorial NW–SE directions (B). The latter component is comparable to the characteristic component observed previously in a preliminary study of Upper Devonian pisolitic ironstones from the same thrust pile, then interpreted as a primary magnetization component (Klootwijk and Conaghan, 1979). However, the presence and uniform direction of this component throughout all the sampled formations in several different thrust sheets proves its secondary origin. Consequently, this component has no bearing on the disputed "Laurasian"-versus-Gondwanan palaeogeographic affinity of the Chitral region. Its equatorial direction is characteristic of a suite of secondary components observed both south and north of the Indus–Tsangpo Suture zone and also south of the western equivalent of this suture zone in northern Pakistan, and which are attributed to initial India–Asia contact. Comparison of these equatorial magnetizations with palaeomagnetic control on the northward movement of the Indian plate obtained from recent ODP Leg 121 studies on the Ninetyeast Ridge, and with recently refined estimates of Greater India's northern extent indicates that initial India–Asia contact was established at, or before, the KTB at equatorial-to-low-northern palaeolatitudes. Comparison of the NW–SE declinations of this secondary component with recent versions of the Indian and Eurasian APWP's indicate counterclockwise rotations of the sampled thrust sheets through 10–30° with respect to the Indian subcontinent and through 60–70° with respect to Eurasia. Some of the recent field components (A) show comparable counterclockwise rotations and indicate the young, latest Cainozoic, ongoing nature of the tectonic activity that led to the Hindukush–Pamir–Karakorum syntaxial bend.

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